VIT-AP UNIVERSITY

DIGITAL IMAGE PROCESSING

(Course Code: CSE 4007)

Module-2:Lecture-3

SPATIAL DOMAIN FILTERING

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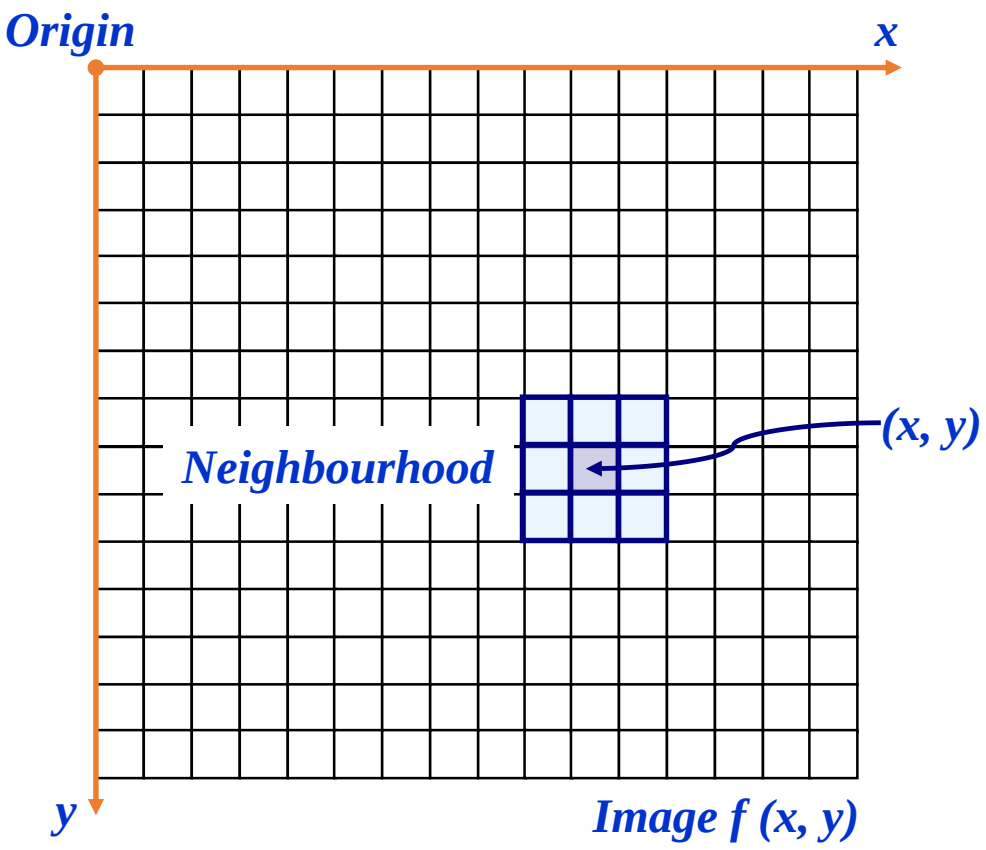
Contents

In this lecture we will look at spatial filtering techniques:

- Neighbourhood operations
- What is spatial filtering?
- Correlation and convolution
- Smoothing operations
- What happens at the edges?
- Sharpening filters
- Combining filtering techniques

Neighborhood Operations

- ✓ Neighborhood operations simply operate on a larger neighborhood of pixels than point operations
- ✓ Neighborhoods are mostly a rectangle around a central pixel
- ✓ Any size rectangle and any shape filter are possible

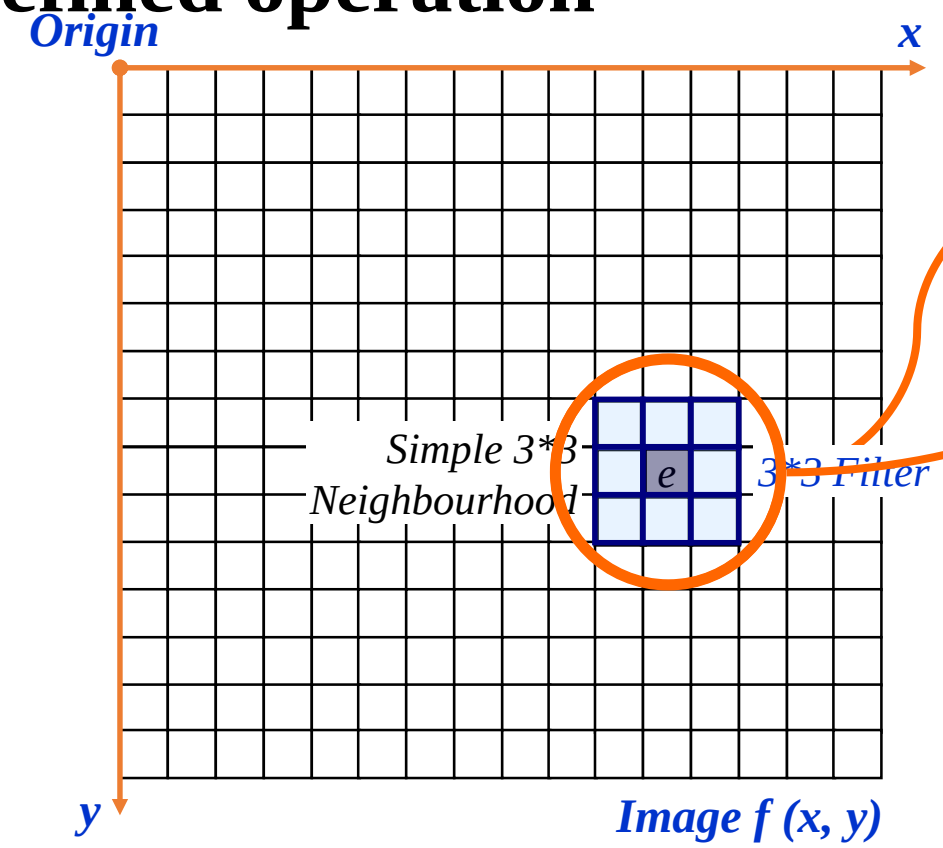


Simple Neighborhood Operations

- ✓ **Some simple neighborhood operations include:**
  - ✓ **Min:** Set the pixel value to the minimum in the neighborhood
  - ✓ **Max:** Set the pixel value to the maximum in the neighborhood
  - ✓ **Median:** The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median). Sometimes the median works better than the average

SPATIAL FILTERING

✓ A spatial filter consists of **(a) a neighborhood**, and **(b) a predefined operation**



a	*b*	*c*
d	*e*	*f*
g	*h*	*i*

Original Image Pixels

*

r	*s*	*t*
u	*v*	*w*
x	*y*	*z*

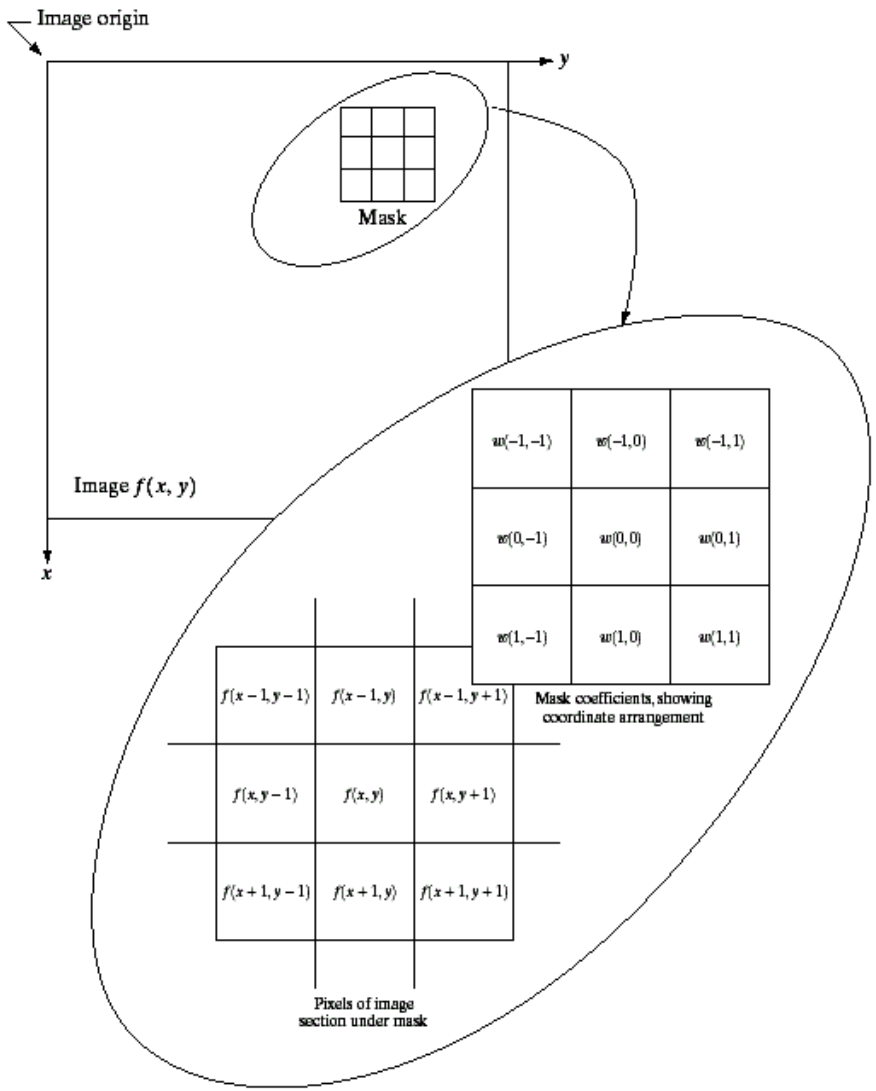
Filter

$$e_{processed} = v*e + r*a + s*b + t*c + u*d + w*f + x*g + y*h + z*i$$

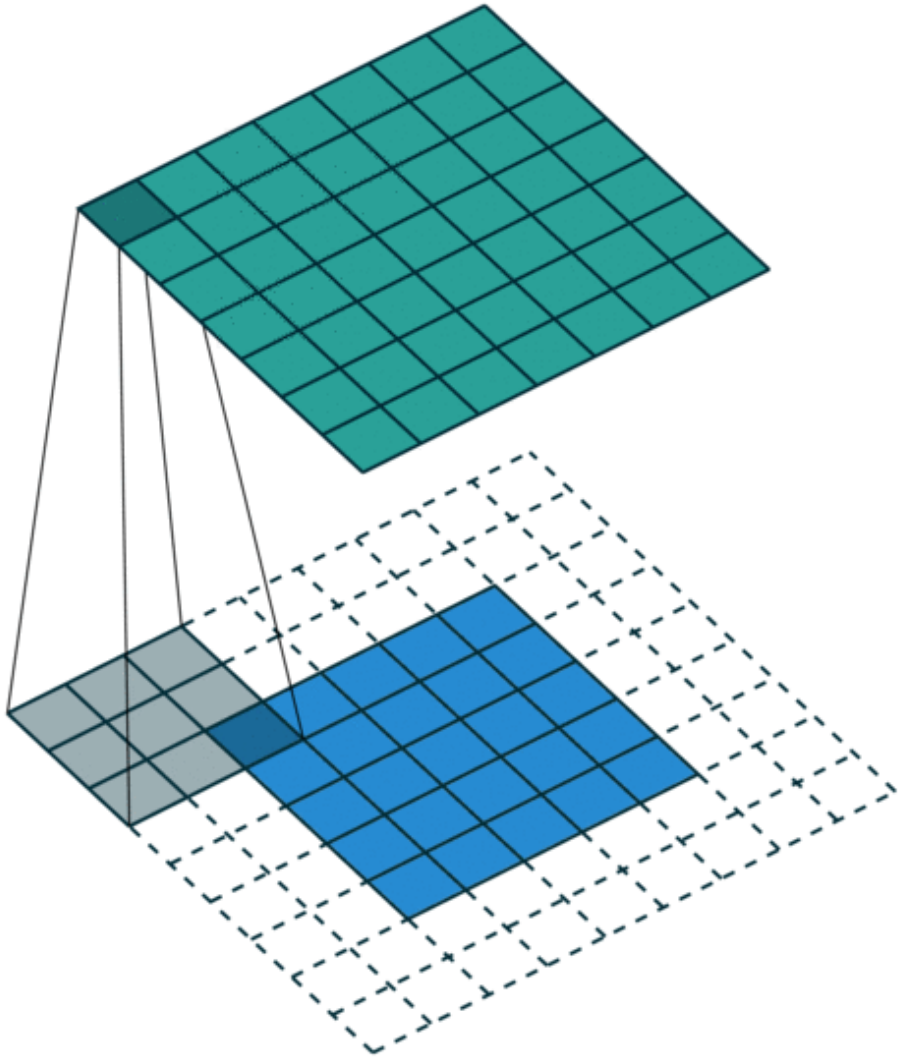
SPATIAL FILTERING-EQUATION FORM

- ✓ The above is repeated for every pixel in the original image to generate the filtered image.
- Filtering can be given in equation form as shown below
- Notations are based on the image shown to the left

$$g(x,y) = \sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$$



2D- FILTERING



1x1	1x0	1x1	0	0
0x0	1x1	1x0	1	0
0x1	0x0	1x1	1	1
0	0	1	1	0
0	1	1	0	0

4		

SPATIAL CORRELATION

The correlation of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \star f(x, y)$

$$w(x, y) \star f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x+s, y+t)$$

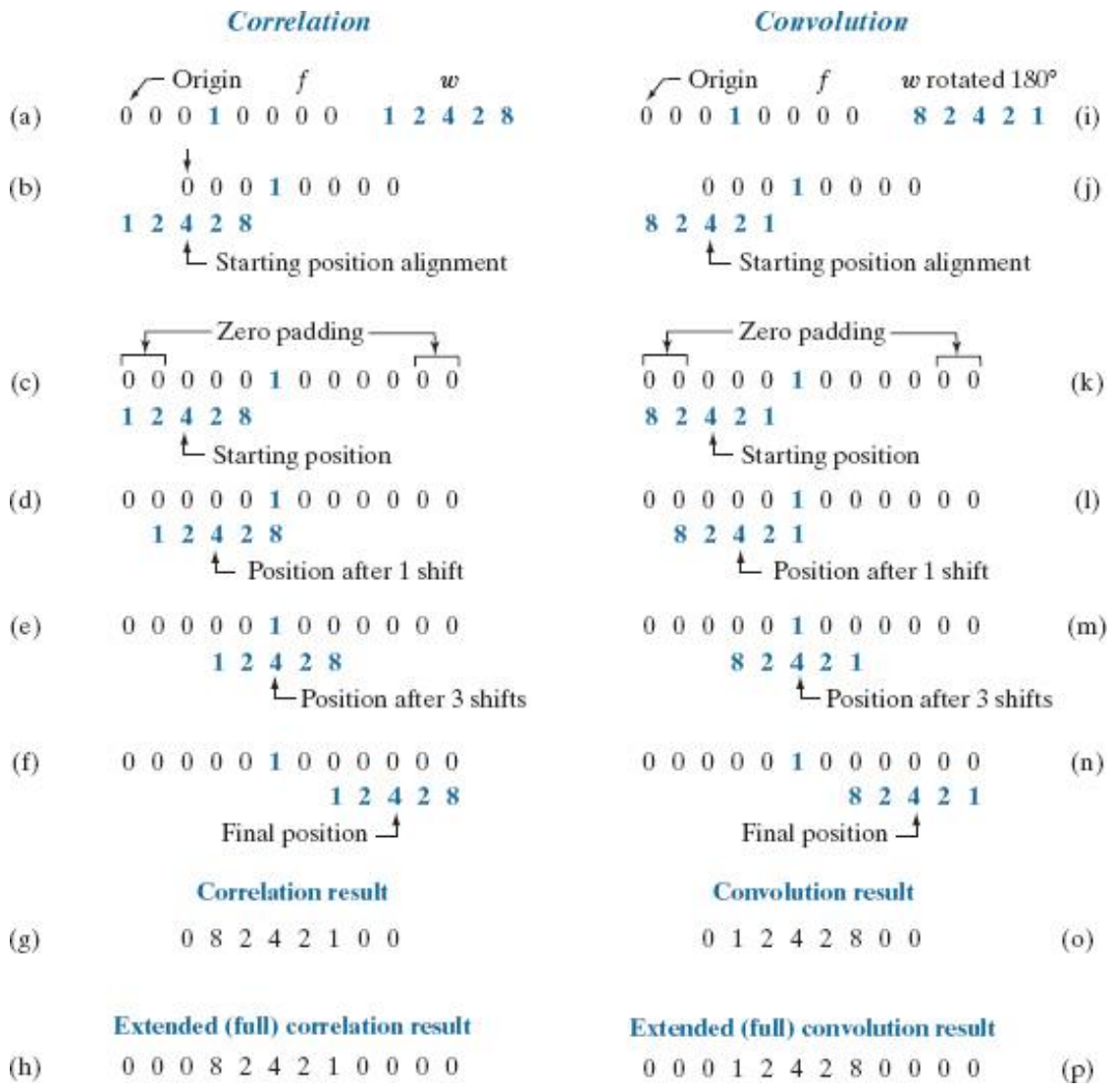
SPATIAL CONVOLUTION

The convolution of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \star f(x, y)$

$$w(x, y) \star f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x-s, y-t)$$

CORRELATION & CONVOLUTION-1D

Illustration of 1-D correlation and convolution of a kernel, w, with a function f consisting of a discrete unit impulse. Note that correlation and convolution are functions of the variable x, which acts to **displace** one function with respect to the other. For the extended correlation and convolution results, the starting configuration places the rightmost element of the kernel to be coincident with the origin of f. Additional padding must be used.



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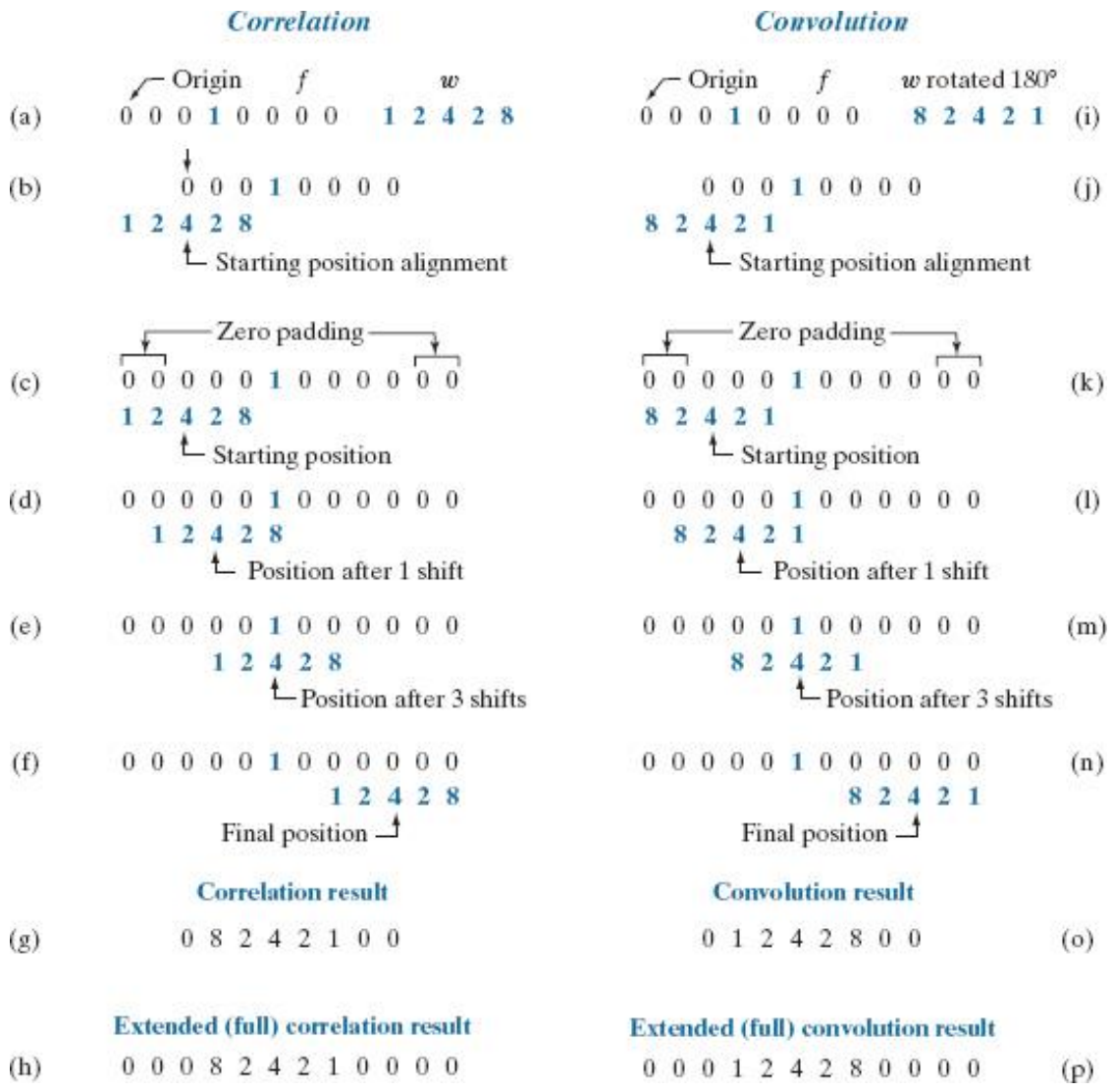

2D- CORRELATION & CONVOLUTION

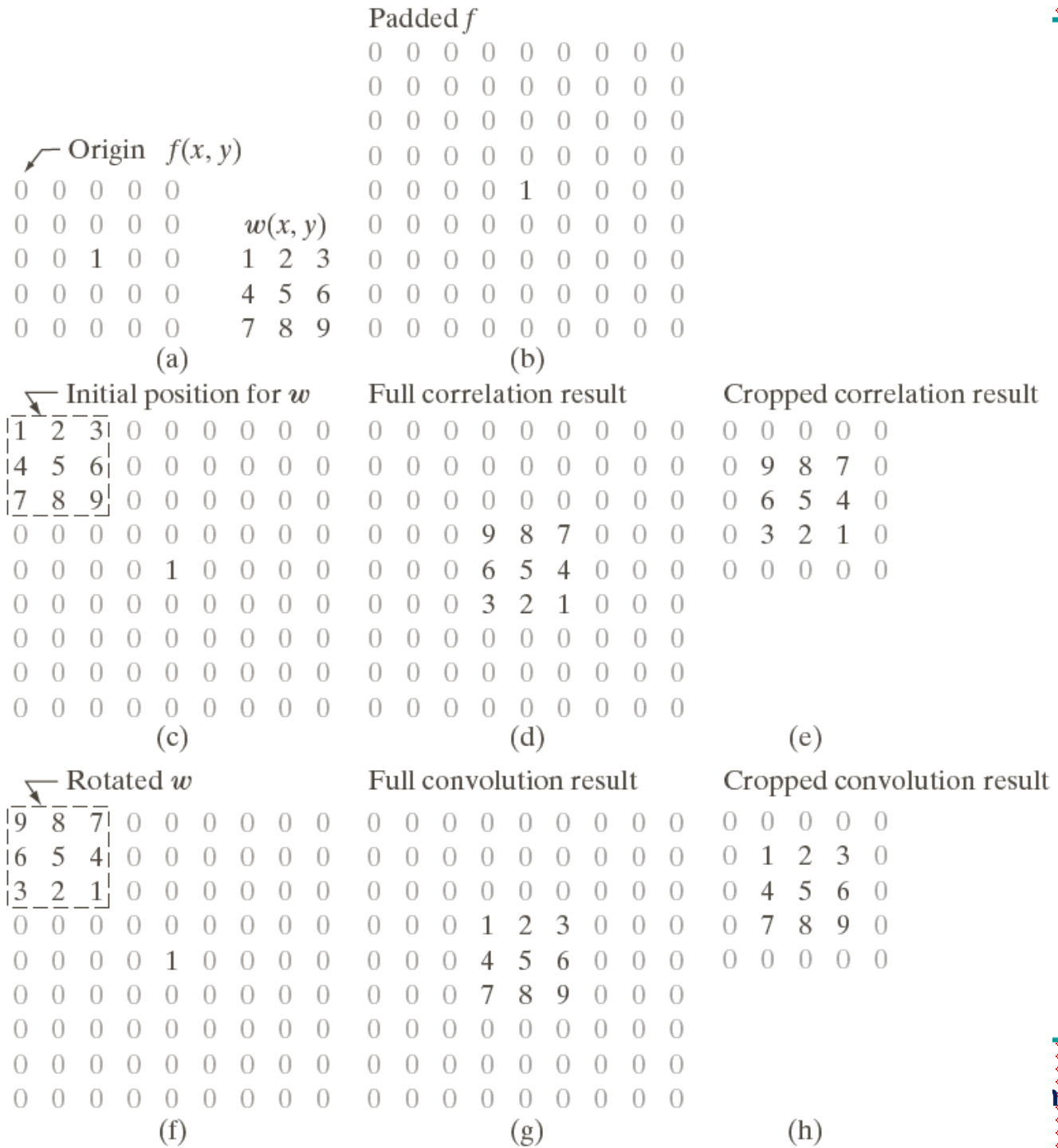

FIGURE 3.30 Correlation (middle row) and convolution (last row) of a 2-D filter with a 2-D discrete, unit impulse. The 0s are shown in gray to simplify visual analysis.

Thank You